



Final Assessment Test (FAT) - JUNE/JULY 2023

Programme	B.Tech.	Semester	Winter Semester 2022-23
Course Title	ELECTRONIC MATERIALS AND DEVICES	Course Code	BECE201L
Faculty Name	Prof. Rahul Narasimhan A	Slot	E2+TE2
		Class Nbr	CH2022232300108
Time	3 Hours	Max. Marks	100

PART- A (6 X 5 Marks)

Answer All questions

01. Calculate the drift mobility and the mean scattering time of conduction electrons in copper at room temperature, given that the conductivity of copper is $5.9 \times 10^5 \Omega^{-1} \text{ cm}^{-1}$. The density of copper is 8.96 g cm^{-3} and its atomic mass (M_w) is 63.5 g mol^{-1} . If the mean speed of the conduction electrons in Cu is roughly $1.6 \times 10^6 \text{ m s}^{-1}$, what is the mean free path between collisions? Note that one mole of copper has N_A (Avogadro's Number) = $6.02 \times 10^{23} \text{ mol}^{-1}$ number of atoms. [5]
02. The density of quantum states per unit volume of the crystal as a function of energy for a free electron is given by $g(E) = 4\pi(2m)^{3/2} h^{-3} (E)^{1/2}$. Calculate the density of states per unit volume with energies between 0 and 1 eV. ($m = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.626 \times 10^{-34} \text{ J s}$) [5]
03. An n-type silicon sample contains a donor concentration of $N_D = 10^{16} \text{ cm}^{-3}$. The minority carrier hole lifetime is found to be $\tau_{p0} = 20 \mu\text{s}$. Assume $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. [5]
- What is the lifetime of the majority carrier electrons?
 - Determine the thermal-equilibrium generation rate for electrons and holes in this material.
04. The reverse-biased generation current is caused by the generation of electrons and holes in the space charge region, in addition to the ideal reverse-biased saturation current. Determine the relative magnitudes (ratio) of the ideal reverse-saturation current density (J_s) and the generation current density (J_{gen}) in a reverse-biased pn junction with voltage V_R at $T=300 \text{ K}$ constituting a depletion region of width W . Use the required parameters from below for your calculations: [5]

$N_A = N_D = 10^{16} \text{ cm}^{-3}$	$\tau_0 = \tau_{p0} = \tau_{n0} = 5 \times 10^{-7} \text{ s}$
$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$	$V_R = 5 \text{ V}$
$D_n = 25 \text{ cm}^2/\text{s}$	$W = 1.214 \times 10^{-4} \text{ cm}$
$D_p = 10 \text{ cm}^2/\text{s}$	$\epsilon_r = 11.7$

05. Determine Q-operating point (V_{CEQ} , I_{CQ}) of the npn transistor circuit shown in Fig.1. Assume $V_{BE} = 0.7$ V and $\beta = 50$. Use $V_{BB} = 4$ V, $V_{CC} = 18$ V, $R_B = 17.4$ k Ω , $R_C = 5.6$ k Ω , and $R_E = 1.2$ k Ω . [5]

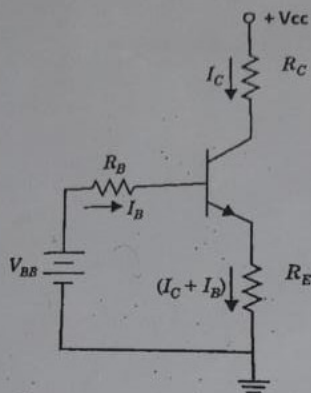


Fig.1

06. Draw and explain the band diagram of an n-channel MOSFET in accumulation, depletion and inversion layer at thermal equilibrium. [5]

PART - B (7 X 10 Marks)

Answer All questions

07. In a Hall effect experiment, a silver slab is placed in a magnetic field. The magnetic field is perpendicular to the largest surface of the slab. The free electron concentration in silver block is $5.85 \times 10^{28} \text{ m}^{-3}$. The current density through the cross section of the slab is $4.4 \times 10^7 \text{ Am}^{-2}$ and it is perpendicular to the magnetic field. The magnetic field applied is $B = 0.95$ Tesla. Answer the following: [10]
- Calculate the Hall electric field.
 - Calculate the Hall coefficient.
 - A brief discussion on any one of the applications of Hall effect device.
08. Explain the energy band structure (E-k) in semiconductor materials. Using E-k diagrams, distinguish between direct and indirect band gap semiconductors. Explain the band-to-band absorption and recombination processes in both direct and indirect band gap semiconductors using E-k diagrams. [10]
09. The total current in a semiconductor is constant and equal to $J = -10 \text{ A/cm}^2$. The total current is composed of a hole drift current and electron diffusion current. Assume that the hole concentration is a constant and equal to 10^{16} cm^{-3} and assume that the electron concentration is given by $n(x) = 2 \times 10^{15} e^{-x/L} \text{ cm}^{-3}$ where $L = 15 \mu\text{m}$. The electron diffusion coefficient is $D_n = 27 \text{ cm}^2/\text{s}$ and the hole mobility is $\mu_p = 420 \text{ cm}^2/\text{Vs}$. Calculate the following: [10]
- the electron diffusion current density for $x = 50 \mu\text{m}$.
 - the hole drift current density for $x = 50 \mu\text{m}$.

10. Consider a silicon pn junction with doping concentrations of N_a and N_d at $T=300\text{ K}$. The intrinsic concentration of silicon at this temperature is n_i . Let the cross-sectional area of the pn junction be A , then calculate the total junction capacitance of this device when a reverse bias of V_R is applied. Note, the dielectric constant of silicon is ϵ_r and permittivity of free space is ϵ_0 . Use the values as given below: [10]

$N_a = 10^{16}\text{ cm}^{-3}$	$n_i = 1.5 \times 10^{10}\text{ cm}^{-3}$
$N_d = 10^{15}\text{ cm}^{-3}$	$V_R = 5\text{ V}$
$\epsilon_r = 11.7$	$A = 10^{-4}\text{ cm}^2$
$\epsilon_0 = 8.854 \times 10^{-12}\text{ Fm}^{-1}$	$e = 1.6 \times 10^{-19}\text{ C}$
$k = 1.3807 \times 10^{-23}\text{ JK}^{-1}$	

11. Referring to the biasing circuit of an npn transistor shown in Fig. 2. Assume $V_{BE} = 0.7\text{ V}$. [10]
- (a) Determine V_{CE} and I_C for $\beta = 50$.
- (b) If the base resistance is increased to $1\text{ M}\Omega$. What happens to the V_{CE} and I_C for $\beta = 100$?

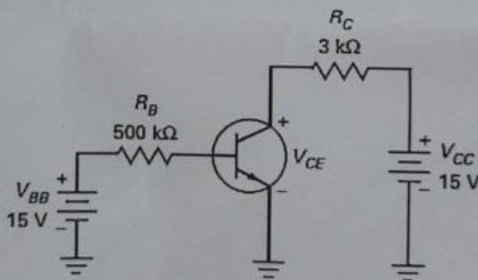


Fig.2

12. Discuss the construction and working of an n-channel depletion-type MOSFET and sketch the transfer characteristics with $I_{DSS} = 10\text{ mA}$ and $V_P = -4\text{ V}$. [10]
13. Explain the properties of carbon nanotube (CNT) and discuss in detail about the methods to grow/synthesize CNT. [10]



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